php

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Lecture4

PHP is a powerful server-side scripting language for creating dynamic and interactive websites.

PHP is the widely-used, free, and efficient alternative to competitors such as Microsoft's ASP. PHP is perfectly suited for Web development and can be embedded directly into the HTML code.

The PHP syntax is very similar to Perl and C. PHP is often used together with Apache (web server) on various operating systems. It also supports ISAPI and can be used with Microsoft's IIS on Windows.

PHP References

At W3Schools you will find complete references of all PHP functions:

- Array functions
- Calendar functions
- Date functions
- Directory functions
- Error functions
- Filesystem functions
- Filter functions
- FTP functions
- HTTP functions
- LibXML functions
- Mail functions
- Math functions

Introduction to PHP

PHP is a server-side scripting language.

What You Should Already Know

Before you continue you should have a basic understanding of the following:

- HTML
- Some scripting knowledge

What is PHP?

- PHP stands for PHP: **H**ypertext **P**reprocessor
- PHP is a server-side scripting language, like ASP
- PHP scripts are executed on the server
- PHP supports many databases (MySQL, Informix, Oracle, Sybase, Solid, PostgreSQL, Generic ODBC, etc.)
- PHP is an open source software
- PHP is free to download and use

What is a PHP File?

- PHP files can contain text, HTML tags and scripts
- PHP files are returned to the browser as plain HTML
- PHP files have a file extension of ".php", ".php3", or ".phtml"

What is MySQL?

- MySQL is a database server
- MySQL is ideal for both small and large applications
- MySQL supports standard SQL
- MySQL compiles on a number of platforms
- MySQL is free to download and use

PHP + MySQL

- PHP combined with MySQL are cross-platform (you can develop in Windows and serve on a Unix platform)

Why PHP?

- PHP runs on different platforms (Windows, Linux, Unix, etc.)
- PHP is compatible with almost all servers used today (Apache, IIS, etc.)
- PHP is FREE to download from the official PHP resource: www.php.net
- PHP is easy to learn and runs efficiently on the server side

Where to Start?

To get access to a web server with PHP support, you can:

- Install Apache (or IIS) on your own server, install PHP, and MySQL
- Or find a web hosting plan with PHP and MySQL support

PHP Installation

What do You Need?

If your server supports PHP you don't need to do anything. Just create some .php files in your web directory, and the server will parse them for you. Because it is free, most web hosts offer PHP support.

However, if your server does not support PHP, you must install PHP.

Here is a link to a good tutorial from PHP.net on how to install PHP5: http://www.php.net/manual/en/install.php

Download PHP

Download PHP for free here: http://www.php.net/downloads.php

Download MySQL Database

Download MySQL for free here: http://www.mysql.com/downloads/index.html

Download Apache Server

Download Apache for free here: http://httpd.apache.org/download.cgi

PHP Syntax

PHP code is executed on the server, and the plain HTML result is sent to the browser.

Basic PHP Syntax

A PHP scripting block always starts with **<?php** and ends with **?>**. A PHP scripting block can be placed anywhere in the document.

On servers with shorthand support enabled you can start a scripting block with <? and end with ?>.

For maximum compatibility, we recommend that you use the standard form (<?php) rather than the shorthand form.

```
<?php
?>
```

A PHP file normally contains HTML tags, just like an HTML file, and some PHP scripting code.

Below, we have an example of a simple PHP script which sends the text "Hello World" to the browser:

```
<html>
<body>
<?php
echo "Hello World";
?>
</body>
</html>
```

Each code line in PHP must end with a semicolon. The semicolon is a separator and is used to distinguish one set of instructions from another.

There are two basic statements to output text with PHP: **echo** and **print**. In the example above we have used the echo statement to output the text "Hello World".

Note: The file must have the .php extension. If the file has a .html extension, the PHP code will not be executed.

Comments in PHP

In PHP, we use // to make a single-line comment or /* and */ to make a large comment block.

```
<html>
<body>
<?php
//This is a comment
/*
This is
a comment
block
*/
?>
</body>
</html>
```

PHP Variables

Variables are used for storing values, such as numbers, strings or function results, so that they can be used many times in a script.

Variables in PHP

Variables are used for storing a values, like text strings, numbers or arrays.

When a variable is set it can be used over and over again in your script

All variables in PHP start with a $ sign symbol.

The correct way of setting a variable in PHP:

```
$var_name = value;
```

New PHP programmers often forget the $ sign at the beginning of the variable. In that case it will not work.

Let's try creating a variable with a string, and a variable with a number:

```php
<?php
$txt = "Hello World!";
$number = 16;
?>
```

PHP is a Loosely Typed Language

In PHP a variable does not need to be declared before being set.

In the example above, you see that you do not have to tell PHP which data type the variable is.

PHP automatically converts the variable to the correct data type, depending on how they are set.

In a strongly typed programming language, you have to declare (define) the type and name of the variable before using it.

In PHP the variable is declared automatically when you use it.

Variable Naming Rules

- A variable name must start with a letter or an underscore "_"
- A variable name can only contain alpha-numeric characters and underscores (a-z, A-Z, 0-9, and _)
- A variable name should not contain spaces. If a variable name is more than one word, it should be separated with underscore ($my_string), or with capitalization ($myString)

PHP String

A string variable is used to store and manipulate a piece of text.

Strings in PHP

String variables are used for values that contains character strings.

In this tutorial we are going to look at some of the most common functions and operators used to manipulate strings in PHP.

After we create a string we can manipulate it. A string can be used directly in a function or it can be stored in a variable.

Below, the PHP script assigns the string "Hello World" to a string variable called $txt:

```
<?php
$txt="Hello World";
echo $txt;
?>
```

The output of the code above will be:

```
Hello World
```

The Concatenation Operator

There is only one string operator in PHP.

The concatenation operator (.) is used to put two string values together.

To concatenate two variables together, use the dot (.) operator:

```
<?php
$txt1="Hello World";
$txt2="1234";
echo $txt1 . " " . $txt2;
?>
```

The output of the code above will be:

```
Hello World 1234
```

If we look at the code above you see that we used the concatenation operator two times. This is because we had to insert a third string.

Between the two string variables we added a string with a single character, an empty space, to separate the two variables.

Using the strlen() function

The strlen() function is used to find the length of a string.

Let's find the length of our string "Hello world!":

```
<?php
echo strlen("Hello world!");
```

```
?>
```

The output of the code above will be:

```
12
```

The length of a string is often used in loops or other functions, when it is important to know when the string ends. (i.e. in a loop, we would want to stop the loop after the last character in the string)

Using the strpos() function

The strpos() function is used to search for a string or character within a string.

If a match is found in the string, this function will return the position of the first match. If no match is found, it will return FALSE.

Let's see if we can find the string "world" in our string:

```
<?php
echo strpos("Hello world!","world");
?>
```

The output of the code above will be:

```
6
```

As you see the position of the string "world" in our string is position 6. The reason that it is 6, and not 7, is that the first position in the string is 0, and not 1.

- What is PHP
- Why PHP
- Where to start
- What is the basic php syntax
- Write a php code that can print a welcome message
- What is the PHP statement for text output
- How to write a comment in PHP
- What does loosely typed language mean What are the rules for variable name?
- What are the functions of a concatenator operator

- What is the difference between

a = 1

And the

A == 1

- What are the following functions do

Strlen()

Strpos()

H.W

1-Write php code that displays your full name using a variable for each part of your name(first name, second name, last name) and then connect the variables

2-Compare between client-side and server-side development techniques